Serie numeriche

1 Stabilire se le seguenti serie a termini positivi sono convergenti, eventualmente in dipendenza dai parametri reali α, β

1.1) $\displaystyle\sum_{n=1}^{+\infty} \frac{1}{\sqrt[7]{n^8}}$ **1.2)** $\displaystyle\sum_{n=1}^{+\infty} \frac{1+n^2}{n\sqrt{n}}$ **1.3)** $\displaystyle\sum_{n=0}^{+\infty} \frac{1+\sqrt{n}}{1+n^2+\sqrt{n}}$ **1.4)** $\displaystyle\sum_{n=1}^{+\infty} \left(\frac{\pi}{n}\right)^n$ **1.5)** $\displaystyle\sum_{n=1}^{+\infty} \left(1+\frac{1}{n}\right)^n$

1.6) $\displaystyle\sum_{n=1}^{+\infty} \frac{n+\log n}{n+n^2+\log^2 n}$ **1.7)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n+\log^3 n}{5+n^3+n\log n}$ **1.8)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n^2 \arctan(n^3)}{1+n^4 \arctan n}$

1.9) $\displaystyle\sum_{n=2}^{+\infty} \frac{n^5+n^6+1}{n^8-1}$ **1.10)** $\displaystyle\sum_{n=2}^{+\infty} \frac{2n^2+n^2\sin n}{n^3\log^2 n}$ **1.11)** $\displaystyle\sum_{n=2}^{+\infty} \frac{n\sqrt{n}+1+\sqrt{n^5}}{n^4-n^3}$

1.12) $\displaystyle\sum_{n=1}^{+\infty} \frac{n+n\sqrt[3]{n}}{n+n^2\log n}$ **1.13)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n^5-n^4}{n+n^3+n^6\log n}$ **1.14)** $\displaystyle\sum_{n=9}^{+\infty} \frac{4+\sqrt{n}}{n\sqrt{n}\log^2 n}$

1.15) $\displaystyle\sum_{n=5}^{+\infty} \frac{2\sin(\log n)}{n\log^3(\sqrt{n})}$ **1.16)** $\displaystyle\sum_{n=2}^{+\infty} \frac{\sqrt{n^3}+n^2+n\sin^2(n^3)}{\sin(n^{-1})+n^3\sqrt{\log^3 n}}$ **1.17)** $\displaystyle\sum_{n=1}^{+\infty} \frac{\tan(n^{-1})}{1+\log^2 n}$

1.18) $\displaystyle\sum_{n=3}^{+\infty} \frac{2^{-\sqrt{n}}+n^2\arcsin(n^{-3})}{3+\log(2n)}$ **1.19)** $\displaystyle\sum_{n=1}^{+\infty} \frac{[n^2/5]}{1+n^3\log n}$ **1.20)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n+\{\pi n^9\}}{1+[n/7]+19n^2}$

1.21) $\displaystyle\sum_{n=2}^{+\infty} \frac{2+\sin(e^n)\cos(\sqrt{n})}{\sin(\arctan(\log n))+11n\log(n^8)}$ **1.22)** $\displaystyle\sum_{n=1}^{+\infty} \frac{ne^{1/n}-n}{\sqrt[3]{n^{14}}\sin(1/n^3)}$ **1.23)** $\displaystyle\sum_{n=1}^{+\infty} \sin\left(\frac{1}{\sqrt[3]{n^4}}\right)$

1.24) $\displaystyle\sum_{n=1}^{+\infty} \frac{\arctan(\frac{1}{n})\log(\frac{n+1}{n})}{\sqrt{\tan(\frac{1}{n})}}$ **1.25)** $\displaystyle\sum_{n=1}^{+\infty} \frac{1}{n^\alpha\sqrt[n]{n}}$ **1.26)** $\displaystyle\sum_{n=1}^{+\infty} \frac{2^n+e^n+3^n}{n+4^n}$ **1.27)** $\displaystyle\sum_{n=2}^{+\infty} \frac{1}{\sqrt[\alpha]{n}\sqrt[\beta]{\log n}}$

1.28) $\displaystyle\sum_{n=1}^{+\infty} \frac{e^n+2^{-n}+\log(1+e^n)}{n^9\log n+e^n}$ **1.29)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n\log^2(1+n^{-\alpha})}{1+\sqrt{n}}$ **1.30)** $\displaystyle\sum_{n=1}^{+\infty} \frac{1-n\sin(\frac{1}{n})}{n^\alpha}$

1.31) $\displaystyle\sum_{n=1}^{+\infty} \frac{e^n}{n^n}$ **1.32)** $\displaystyle\sum_{n=1}^{+\infty} \frac{e^{n^2}}{n^n}$ **1.33)** $\displaystyle\sum_{n=1}^{+\infty} \frac{e^n}{n!}$ **1.34)** $\displaystyle\sum_{n=1}^{+\infty} \frac{e^{\alpha/n}}{n^\alpha}$ **1.35)** $\displaystyle\sum_{n=1}^{+\infty} \frac{2^{\alpha n}}{n^\alpha}$

1.36) $\displaystyle\sum_{n=1}^{+\infty} \frac{e^{n^2}}{(n!)^n}$ **1.37)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n!}{n^n}$ **1.38)** $\displaystyle\sum_{n=1}^{+\infty} \frac{1}{n^\alpha\,(n!)^{1/n}}$ **1.39)** $\displaystyle\sum_{n=1}^{+\infty} \frac{\log(n!)}{n^\alpha\log^3 n}$ **1.40)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n^{\alpha n}}{(n!)^2}$

2 In caso di convergenza, calcolare la somma delle seguenti serie (in alcuni casi è opportuno ricondursi a serie telescopiche)

2.1) $\displaystyle\sum_{n=0}^{+\infty} \frac{1}{3^n}$ **2.2)** $\displaystyle\sum_{n=0}^{+\infty} \left(\frac{9}{\pi^2}\right)^n$ **2.3)** $\displaystyle\sum_{n=2}^{+\infty} (\log 2)^{-5n}$ **2.4)** $\displaystyle\sum_{n=0}^{+\infty} \left(\frac{\pi}{3}\right)^n$ **2.5)** $\displaystyle\sum_{n=5}^{+\infty} \left(\frac{\pi}{4}\right)^{-\frac{n}{2}}$

2.6) $\displaystyle\sum_{n=1}^{+\infty} \left(\frac{1}{(n+1)!}-\frac{1}{n!}\right)$ **2.7)** $\displaystyle\sum_{n=9}^{+\infty} \frac{1}{n(n+1)}$ **2.8)** $\displaystyle\sum_{n=1}^{+\infty} \frac{1}{(2n-1)(2n+1)}$ **2.9)** $\displaystyle\sum_{n=2}^{+\infty} \frac{\sqrt{n+1}-\sqrt{n}}{\sqrt{n^2+n}}$

2.10) $\displaystyle\sum_{n=0}^{+\infty} \left(e^{-(n+1)}-e^{-n}\right)$ **2.11)** $\displaystyle\sum_{n=1}^{+\infty} \log\frac{n^2+2n}{n^2+2n+1}$ **2.12)** $\displaystyle\sum_{n=12}^{+\infty} \frac{2n+1}{n^2(n+1)^2}$

3 Stabilire se le seguenti serie sono convergenti e se sono assolutamente convergenti

3.1) $\displaystyle\sum_{n=1}^{+\infty} \frac{(-1)^n}{n}$ **3.2)** $\displaystyle\sum_{n=1}^{+\infty} \frac{(-2)^n}{\sqrt{n}}$ **3.3)** $\displaystyle\sum_{n=2}^{+\infty} (-1)^n \frac{1+n}{1+n\sqrt{n}}$ **3.4)** $\displaystyle\sum_{n=0}^{+\infty} (-1)^n \frac{3^n+1}{2^{2n}+1}$

3.5) $\displaystyle\sum_{n=0}^{+\infty} (-1)^n \frac{n^n e^{-n}}{n!}$ **3.6)** $\displaystyle\sum_{n=1}^{+\infty} (-1)^n \sin\left(\frac{1}{n}\right)$ **3.7)** $\displaystyle\sum_{n=1}^{+\infty} (-1)^n \, n \arctan\left(\frac{1}{n^2}\right)$

3.8) $\displaystyle\sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{\sqrt[3]{n}}$ **3.9)** $\displaystyle\sum_{n=1}^{+\infty} \frac{\sin n}{n^2}$ **3.10)** $\displaystyle\sum_{n=12}^{+\infty} \frac{\cos(\pi n)}{\sqrt{n}}$ **3.11)** $\displaystyle\sum_{n=0}^{+\infty} \sin\left(\pi \frac{n^2+n+1}{n+1}\right)$

3.12) $\displaystyle\sum_{n=1}^{+\infty} \frac{(-1)^n \log n}{n}$ **3.13)** $\displaystyle\sum_{n=1}^{+\infty} \frac{(-1)^n}{2n+(-1)^n}$ **3.14)** $\displaystyle\sum_{n=1}^{+\infty} \frac{\cos(\pi n)}{[n]}$

4 Si determini il raggio di convergenza delle seguenti serie di potenze

4.1) $\displaystyle\sum_{n=1}^{+\infty} \frac{x^n}{n^2}$ **4.2)** $\displaystyle\sum_{n=1}^{+\infty} \frac{x^n}{3^{2n}}$ **4.3)** $\displaystyle\sum_{n=1}^{+\infty} (2n(n-1)(n-2)-1)\, x^n$

4.4) $\displaystyle\sum_{n=0}^{+\infty} (\sqrt{n+1}-\sqrt{n})\, x^n$ **4.5)** $\displaystyle\sum_{n=1}^{+\infty} 2^{-\sqrt{n}}\, x^n$ **4.6)** $\displaystyle\sum_{n=1}^{+\infty} \frac{n!}{2^n}\, x^n$

4.7) $\displaystyle\sum_{n=1}^{+\infty} \left(\frac{1}{(n+1)!} - 4^n + 3\right) x^n$ **4.8)** $\displaystyle\sum_{n=4}^{+\infty} \frac{2n+(-1)^n}{(2n)!}\, x^n$